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Distributed Flood Detection Network

LoRa mesh flood detection and mapping system

A network of solar-powered, LoRa-connected sensor nodes deployed in rural or urban areas to measure local water levels and relay readings across a peer-to-peer mesh. A central server aggregates point measurements and cross-references them against a Digital Elevation Model (DEM) to interpolate flood extent and depth in near-real time, targeting clogged drainage systems and overland flooding in areas without traditional gauge infrastructure.

Technical Challenges

- LoRa mesh routing: P2P relay without centralized gateways
- Sustained solar power budget: charging, deep-sleep scheduling, and adaptive TX intervals must keep nodes alive through overcast/winter conditions without mains power.
- Sensor durability: Ultrasonic/pressure sensors exposed to adverse conditions need robust waterproof enclosures while keeping the sensing element exposed.
- Flood extent interpolation: Converting sparse point measurements into a continuous flooding surface over a DEM is sensitive to node density, DEM resolution, and unmapped features.

Skills Needed

- Embedded systems (ESP32/STM32 firmware, low-power design, solar panel).
- RF networking (LoRa, mesh routing, FCC regulatory compliance).
- Geospatial analysis using DEMs
- Backend/visualization (time-series DB, web map dashboard with tile overlay, alerting pipeline).
- Low power digital electronics. Sleep timing, charging, and overall efficiency.

Deliverables / Demo Items

- Working solar-powered sensor node prototype (PCB, enclosure, firmware) with LoRa transmission.
- Multi-node mesh demo relaying data across hops to a base station.
- Web dashboard displaying real-time sensor readings and interpolated flood extent over a map.
- Bill of materials, per-node cost analysis, deployment guide, and field test report (accuracy, range, power budget).

Related Projects

Project	Summary	Difference	Link
FloodNet	NYC ultrasonic street-flood sensors on LoRaWAN/TTN; mains-powered, city gateways.	No mesh, no GIS extent, no solar	https://www.floodnet.nyc/
Meshtastic	Mesh network for LoRa device communication. Irrelevant to flooding but provides groundwork for mesh coms	No flooding	https://meshtastic.org/docs/introduction/
FloodMapper	ESP32 LoRaWAN water-level node for small rivers; single-node firmware, TTN.	Single node only, no GIS	https://www.flood-mapper.com/

Key differentiator: This project jointly integrates (1) gateway-independent LoRa mesh relay, (2) autonomous solar-powered nodes for unattended rural/urban deployment, and (3) DEM-based spatial flood extent interpolation. All of which are capabilities addressed individually but not combined in existing work.